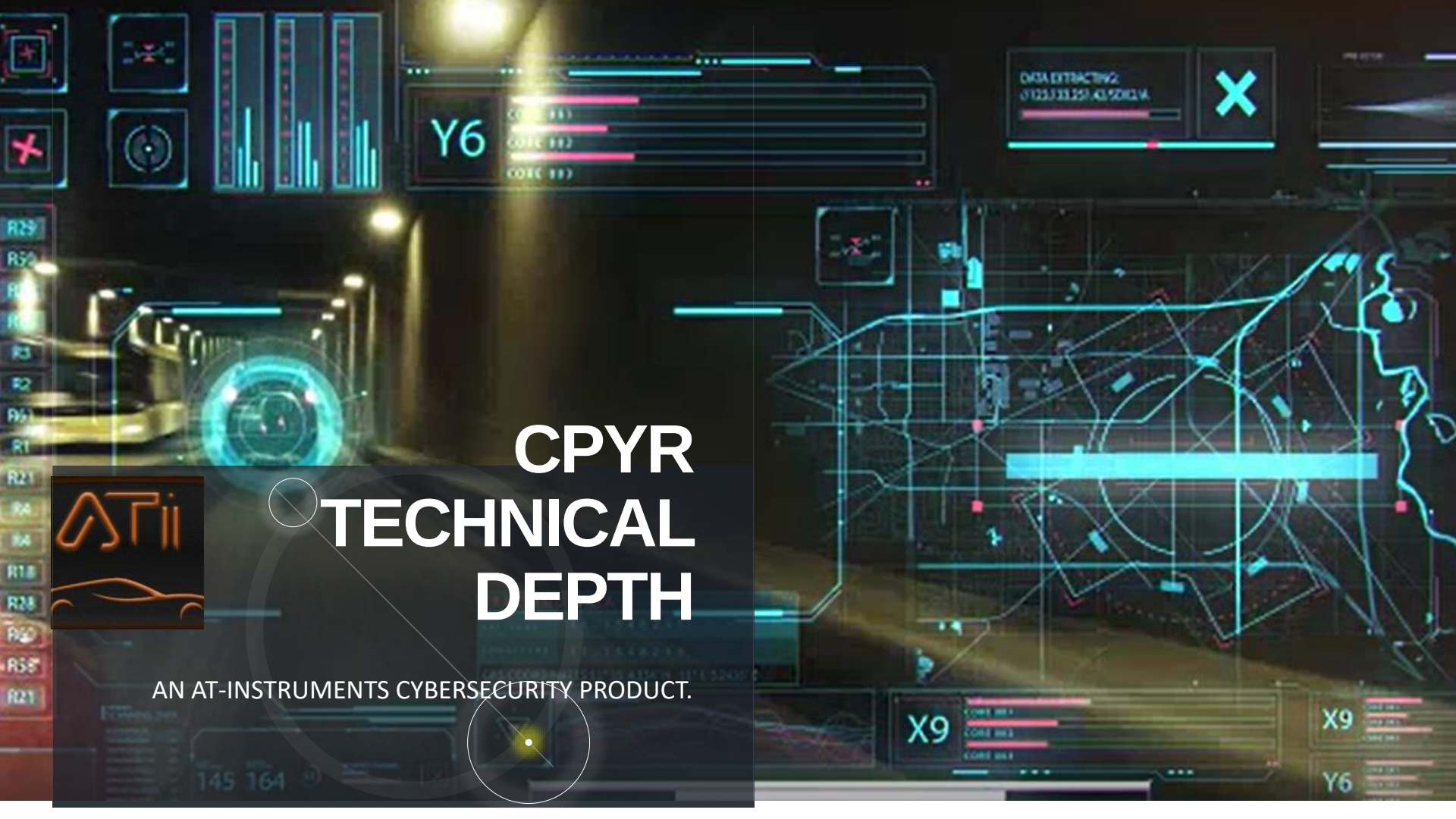




CPYR TECHNICAL DEPTH

AN AT-INSTRUMENTS CYBERSECURITY PRODUCT.



CPYR

Overview

Understanding the problems

Aspects of Automotive Security prevention

Traditional defenses mechanism

Project objective

Continuous learning solution

CPYR Agent

CPYR Box

Project timeline

Overview

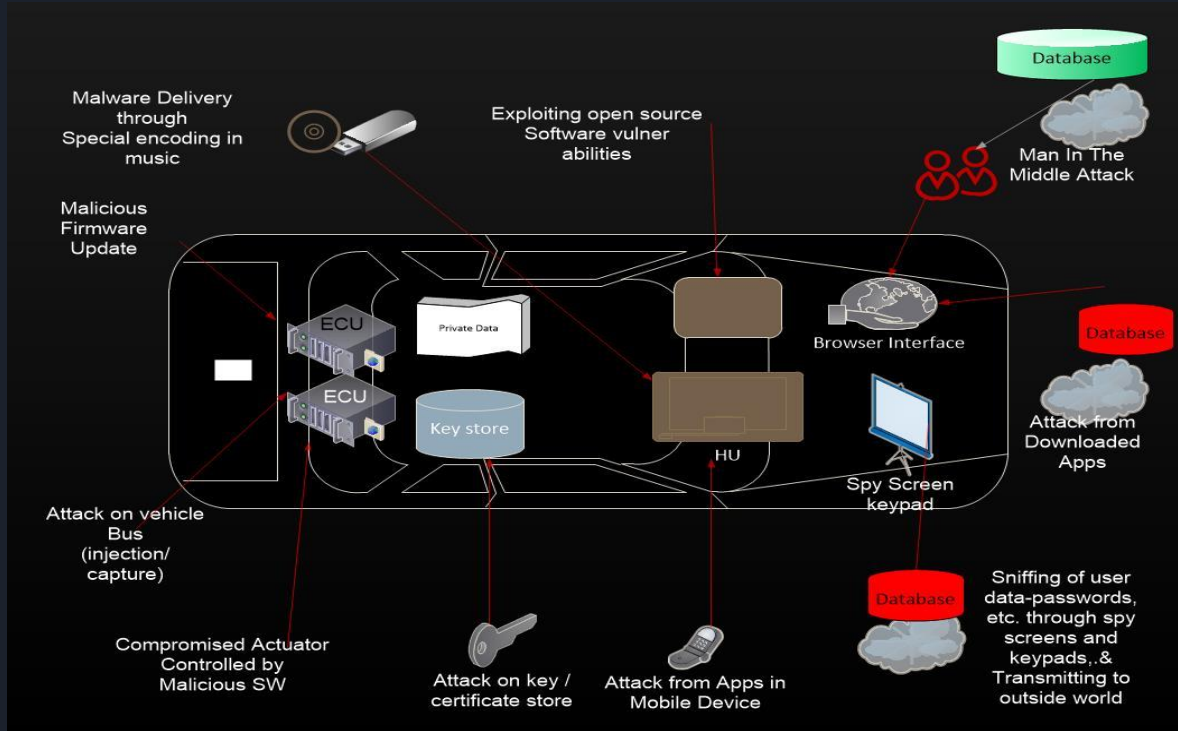
- Autonomous cars are targeted to be released by 2020, and the end objective is passengers fully entrusting there Autonomous vehicles to drive them.
- Autonomous vehicles means Embedded SW that is capable of :
 1. Communicating directly vehicle to vehicle (V2V) and vehicle to infrastructure (V2I) digital wireless communication (collectively referred to as V2X) enables vehicles to communicate with each other and roadside equipment.
 2. Communicating with sensors wired and wirelessly .
 3. Depending on SW updates from the Authentic partners.
- Also Infotainment , OBD, GPS expands the attack surface.
- Cyberattacks are inevitable

Attack surface



Even trivial attack surfaces that the Auto manufacturers could cause extremely life threatening hazards, ex Charlie Miller a security researcher ,and Chris Valasek, director of Vehicle Security Research at IOActive, have exposed the security vulnerabilities in automobiles by hacking into cars remotely, controlling the cars' various controls from the radio volume to the brakes.

The attacker surface expands as the car gets smarter, and hence the protection of the targeted ECU is becoming more efficient than trying to limit the surface of attack.



Aspects of Automotive Security prevention

- Monitoring Inputs to ECU
- Preventing remote Code Executing Cyberattacks.
- Detecting behavior anomalies.

All of these Aspects have to be preserved with two important factors

- In Cybersecurity Speed is Paramount
- Cyber attacks are becoming more creative and more sophisticated to penetrate multiple security layers.

It is very hard (practically impossible) to predict all types of attack vectors evolving throughout time

Examples of Attacks Signatures

- Resources consumption : CPU usage, Memory Heap, Queue high/low water marks, Communications
- Programing errors : Write to holes in memory, Reads from Write-only registers , Write to flash
- Access Violations : Booting not from ROM, Manipulation of device or peripheral, execute code from RAM, ..
- Buffer overflows, Arc injection, Stack execution (code injection)
- Key scanning on bootloaders (Compromising the Secure boot) using Hamming , Shannon,..
- Flashes non integrity, RAM Test executed in abnormal regularity
- WD resets, Over temperatures , ADCs abnormal values
- Undefined startup / shutdown sequences
- OS errors & reports, Traps, Elevation of privileges
- Safety monitors being fired regularly , including SMU alarms, Memory CRCs,
- Using Autosar leaks to reach to a Failsafe state in order to reset the system
- Crafting an attack against the error state within the CAN protocol called "Bus-Off".
- Sensitive data not being erased from RAM.

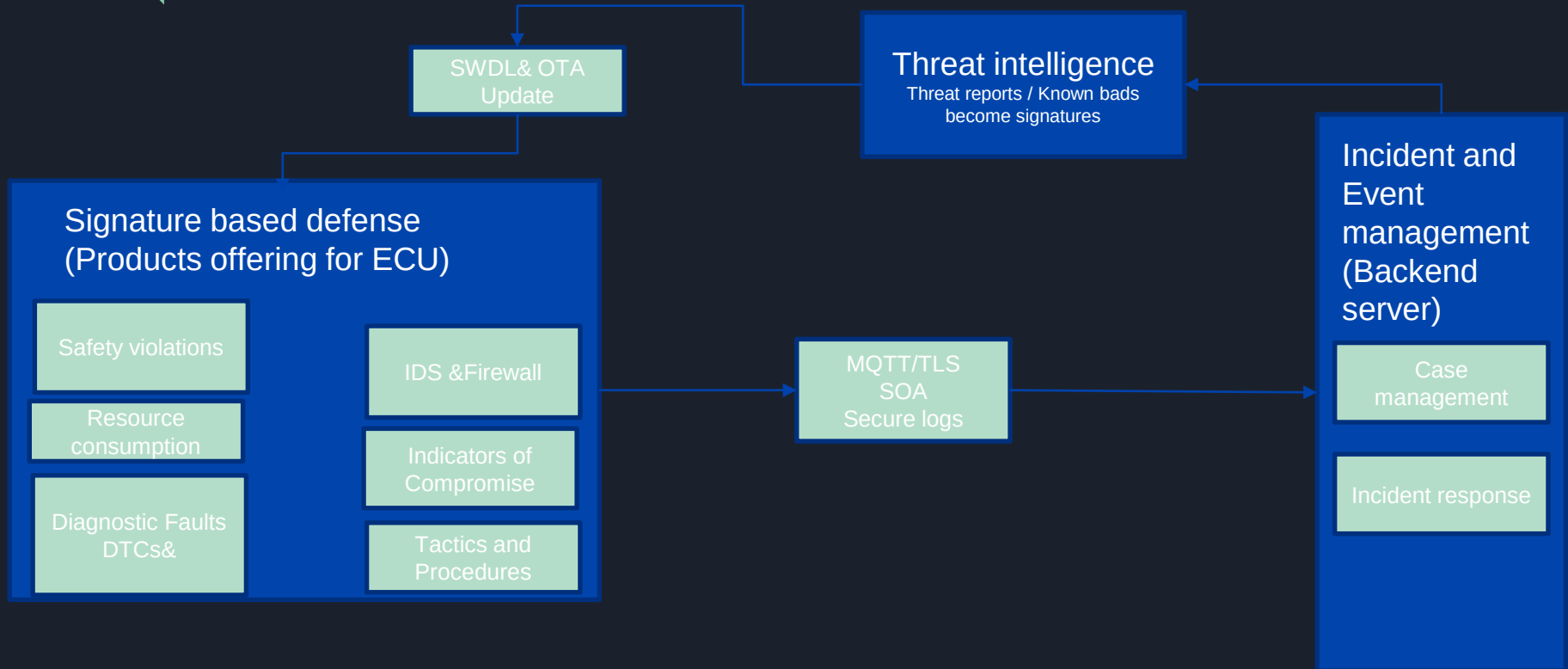
Examples Secure logging

- Memory read failures at file system
- Secure Boot failures MPU exceptions memory access Host
- MPU exceptions memory access Tree Runner
- MQTT Certificate Authentication Failure
- MQTT connection drop off
- SOTA Signature verification failure
- SWDL Signature verification failure
- Secure Diagnostics SEED KEY authentication Failure Backend Authentication failure
- Secure log the data validation failures on CAN for security critical messages
- Securely log the data validation failures on Ethernet for security critical messages
- Attempts to turn on/off the transceivers i.e. communication control service \$28
- Attempts to Clear DTC information \$14

Examples Secure logging

- Certificate validation failures during the rotation
- Attempts for SW installation events
- Attempts to change DTC control settings i.e UDS service \$85
- Attempts to perform Security Access \$27
- Switch Diagnostic failures
- SPI communication failures
- Sensor Blockage detection
- NVM corruption of security critical data
- Stack overflow
- Access control events

Traditional defenses mechanism





Problems of the traditional mechanisms

01

Attack vectors keep evolving and hence continuous analysis of the SW behaviour is required → **That can't keep up with the Adversarial mission pace.**

02

OEM's analysis and performing Software updates requires huge time before preventing the adversaries → **Every second in delays in alerting and detection is more time for adversaries to cause more damage.**

03

Detailed and Cyclic Secure logs for performing adequate anomaly analysis requires a huge Bandwidth and regular updates to the OEM's Backend server → **Huge backend servers are required and very costly communication is required to move gigabytes of logging data.**



Project objective

Detecting the unknown using unsupervised behavior models to expose malicious Network & Sw activities. That learns over time and differentiate adversarial anomalies from Embedded/Random Driver behavior

Deep analytics enrichment to develop tailored behavioral model by leveraging deep packet inspection, AI / ML and SWC intercommunication analysis.

We can build (including the preprocessing) and train deep neural network on a GPU, while performing normal Test drives , where the attack surface is secured (i.e. Secure environment).

Continuous learning solution

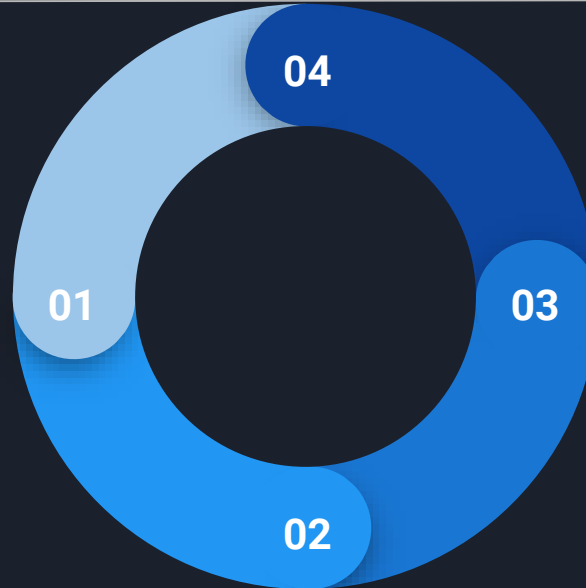
Data Parsing

Parsing the Raw data quickly and staging it in a readable format for the machine learning models is a key factor of layered security model

Network segmentation & Anomaly scoring

AI & Automotive CS experts collaborate to develop trusted network segments.

Anomaly scoring is done with network segment and cyber context minimizing false positives



Segment – Specific Adversarial risk

Engine is fully Embedded SW tailored and remains assumption agnostic without pretraining bias

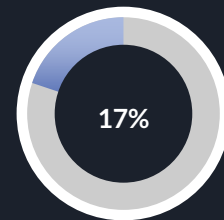
User feedback

Encoded in in reinforcement learning algorithm differentiates adversarial anomalies

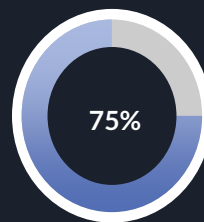
AI-enabled enrichment

We built an adaptive anomaly detection engine fully on GPU that can detect new attack vectors as they are implemented, and designed to mitigate false positives. How?

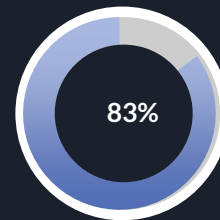
1. As thousands of miles being spent on test drives healthy system behavior of the ECU could be sent by the CPYR Agent to the CPYR box(server). → [Develop a trusted network segment \(vector space\)](#)
2. Agent process some simulated attacks on the ECU and sends the infected system behavior to the CPYR box . → [Develop a n anomaly threshold](#)



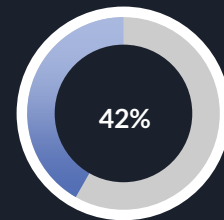
Beaconing with Bayesian endpoint profiling



DGA detection with deep neurals



Interpretable malware detection with a light GBM

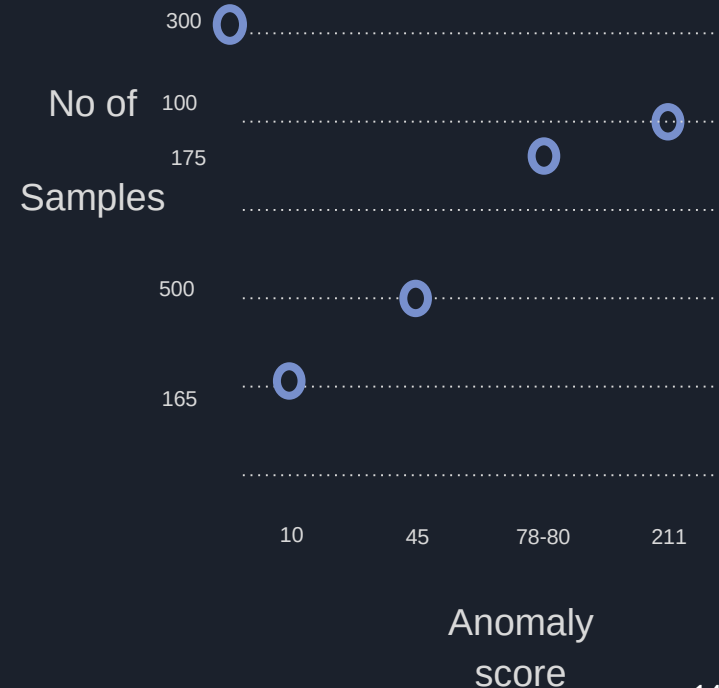


Credential misuse detection with XGboost

Minimizing the false positives

CPYR box adds classifiers for AI attack vector detection → Anomaly scoring is done with network segments and cyber context, minimizing the false positives.

Anomaly Segments	Description
OS Privilege Escalation	Use /sbin/iftwatchd to gain privileges by providing an arbitrary program name as a command-line argument. Gain Root Access to the system Affected: QNX 6.5.0, 6.5.0 SP1, 6.4.1
DoS Comstack	Inject dummy CAN messages with low frame IDs (High priority messages) to the CAN bus with low periodicity
Exfiltration of sensitive data	Analyze dump of MQTT Traffic to check credentials and topics data. Resend the same messages again





Segment –Specific Adversarial risk

CPYR box produces Bayesian behavior model

(With no need for memorizing the behavior throughout the thousands of miles but instead a set of parameters).

This is done through a Streaming Engine composing of

- K-means :could be used to Normalize the features space and builds clusters to the Vector space.
- Composite Nearest Neighbor, scores anomalousness within retrospective windows using distance measures and neighborhood algorithms.

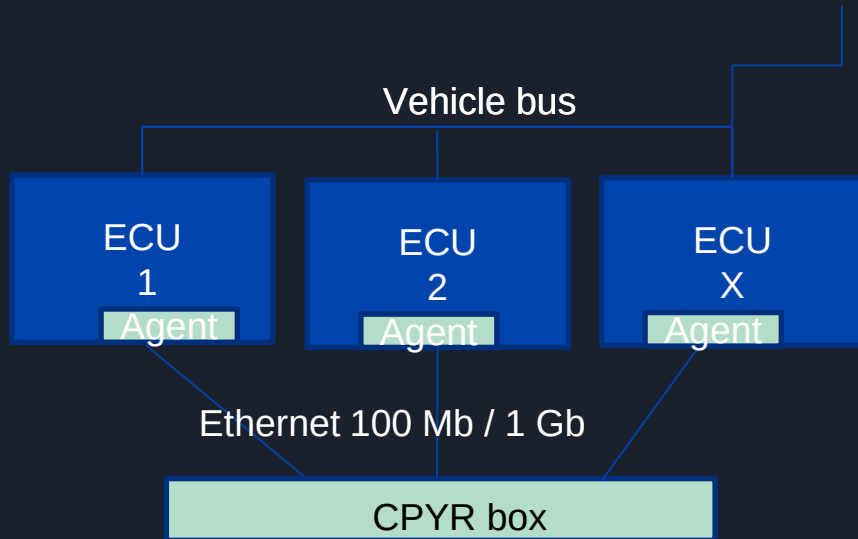


Reinforcement learning

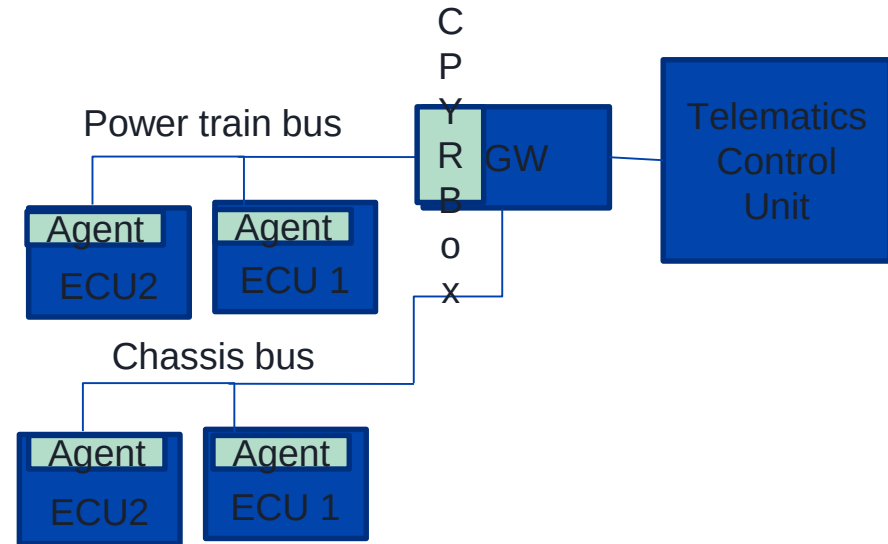
And reinforcement learning with special types of alarms are included to provide continuous learning and improving the vector space.

- Cyber expert evaluate the anomalies alerts , based on it's classification at it's respective segments, and hence the Expert could evaluate if this segment makes sense or should be removed.
- Retraining happens only when you want to re-segment when you have a hard time to allocate new data

CPYR suited for group of ECUs



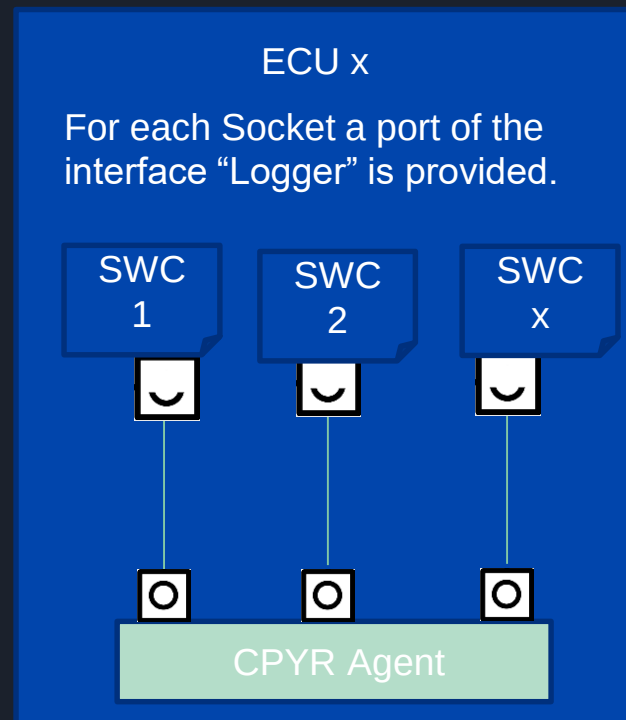
Or for a Complete vehicle anomaly detection



CPYR Agent

All entities connected to this AUTOSAR Runtime Environment (RTE) must comply to the AUTOSAR specification to ensure integration of the user specific software. The AUTOSAR architecture consist of:

- Software components (SWC) for the application
- Basic software components for low-level services
- And a Runtime Environment





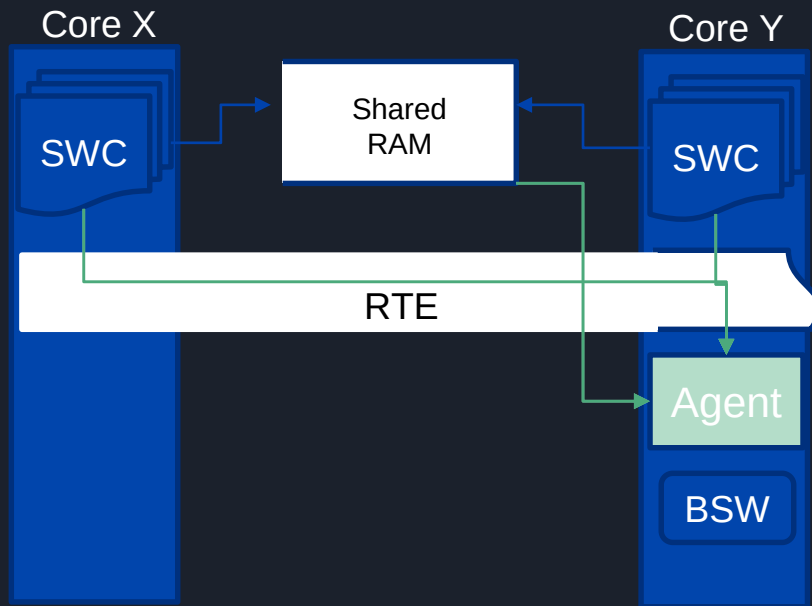
CPYR agent is AUTOSAR compliant

AUTOSAR is a standard that's gaining momentum worldwide and that promotes the standardization and reuse of automotive embedded software, electric and electronic components. AUTOSAR's objectives are modularity and configurability. It defines a layered software architecture allowing integration of electronic control units (ECU) with configurability. And it enables function reuse from different suppliers. Standardized interfaces are defined between each layer, and standardized datatypes are defined.

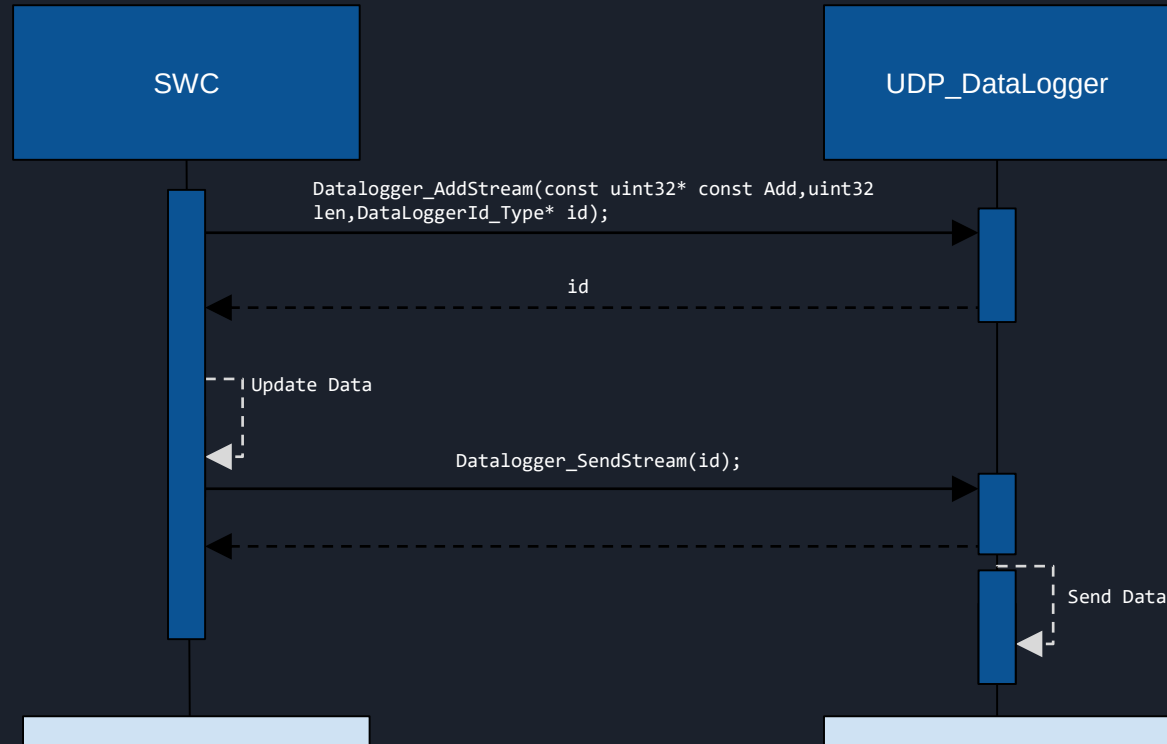


CPYR Agent

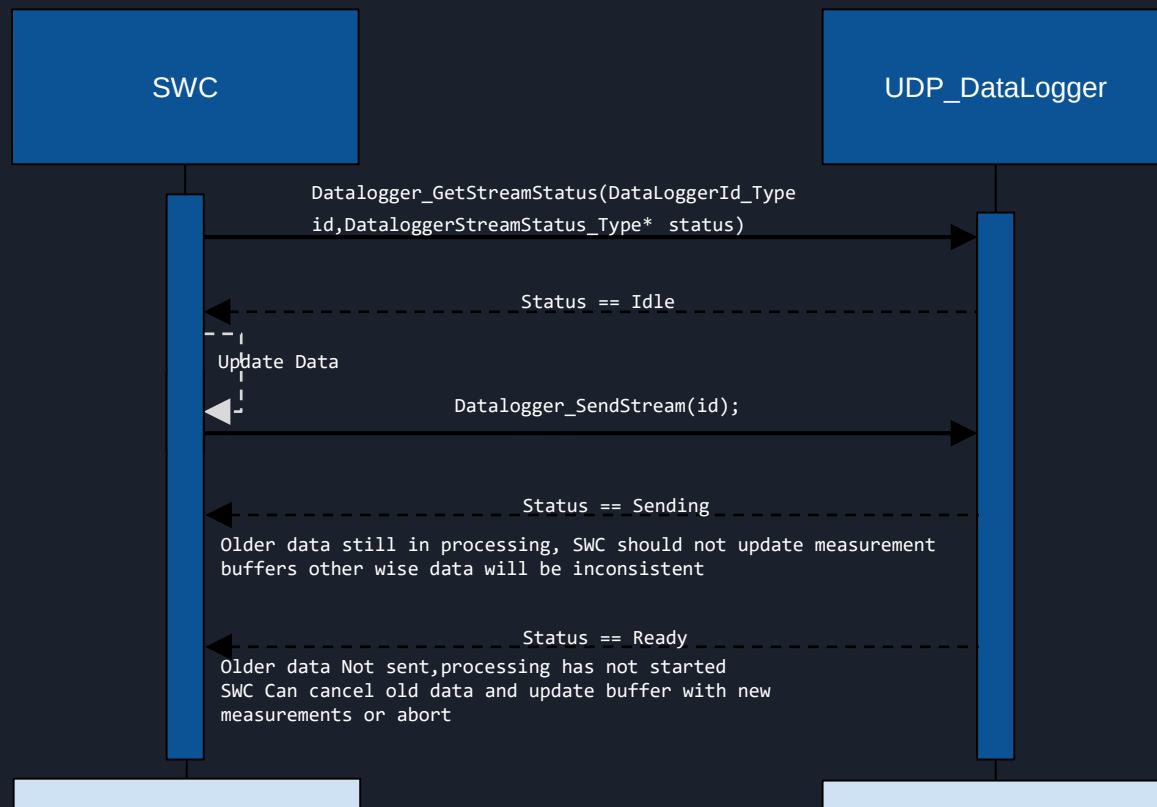
- Provides SWC ability to send UDP Frames over Ethernet through pre-configured sockets (IP and Port combination).
- Pre-Compile configurable number of streams per socket.
- Allocation and formatting of Measurement buffers and data is responsibility of SWCs
- Measurement buffers have to be grouped in a separate RAM section



Call Sequence



Call Sequence

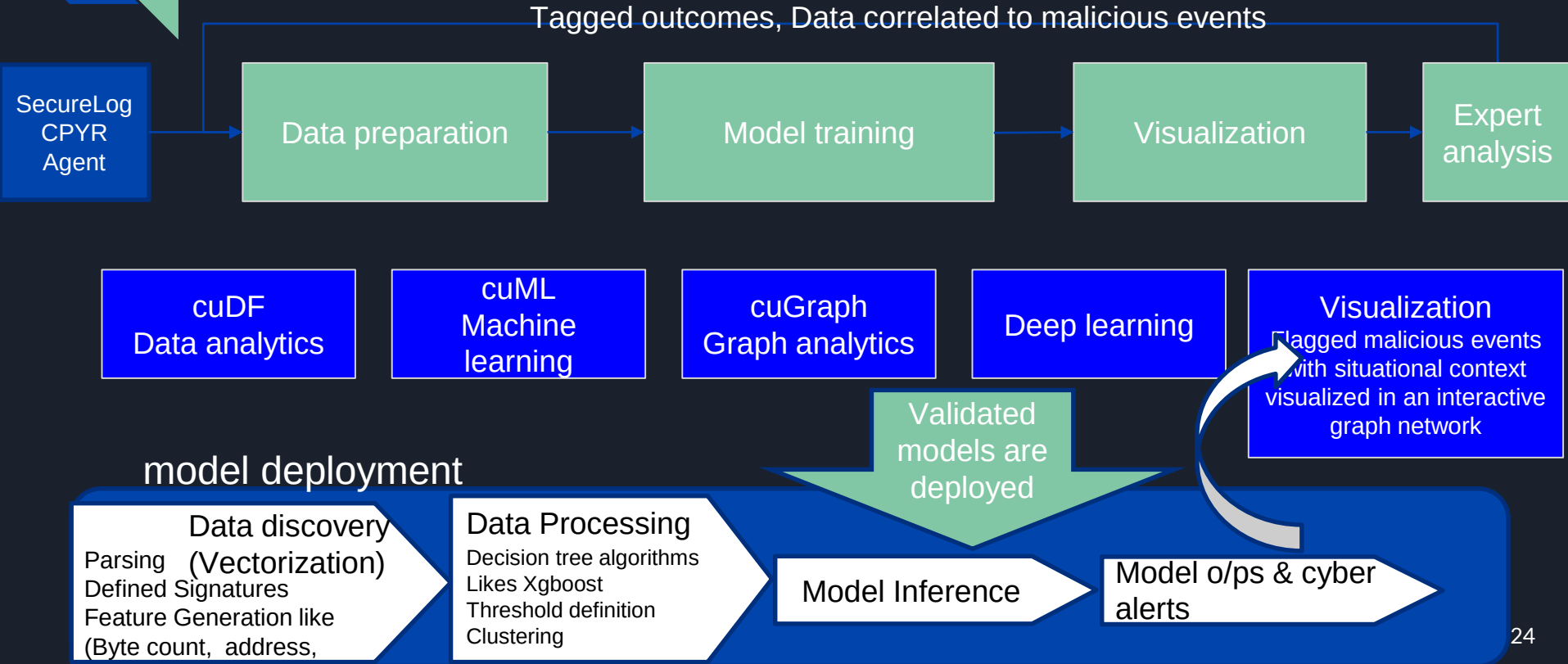




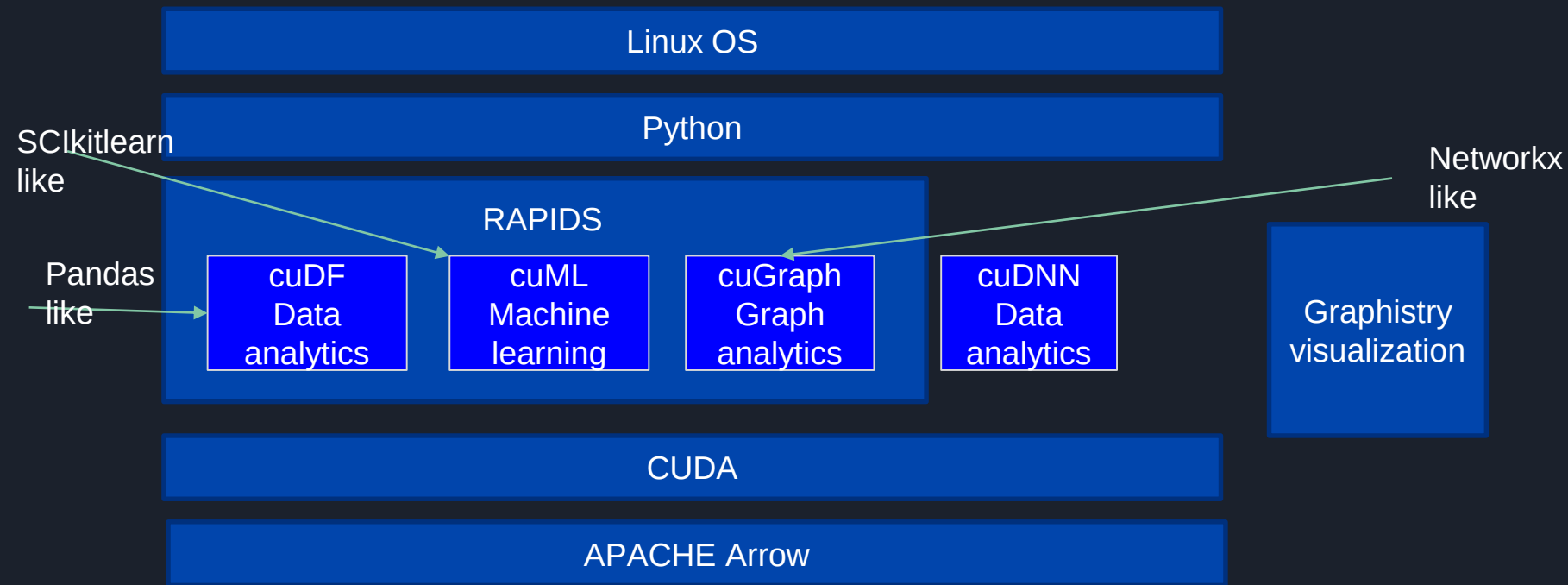
CPYR box

- The CPYR box is a GPU based processing , composing of hundreds of cores that can handle thousand of threads (unlike normal CPUs) , the GPU is optimized carry the same operation over and over effectively
- GPUs are suitable for Cybersecurity analysis, since there is a huge amount of Data and throughput, and the data generated is structured .
- We can build (including the preprocessing) and train deep neural network on a GPU, while performing normal Test drives , where a secure environment is guaranteed
- CPYR box provides the cyber expert a visual behavioral analysis of the Vehicle at the secured test environment versus the anomalous behavior at the Vehicle market deployment .

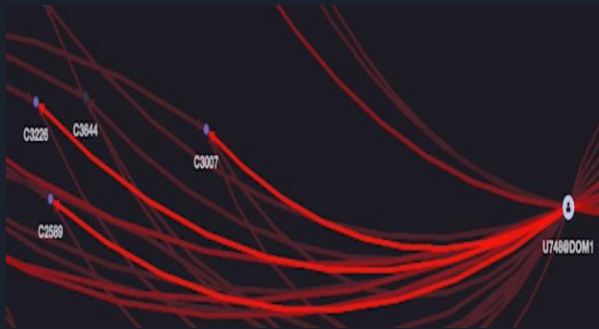
GPU Data Science Pipeline



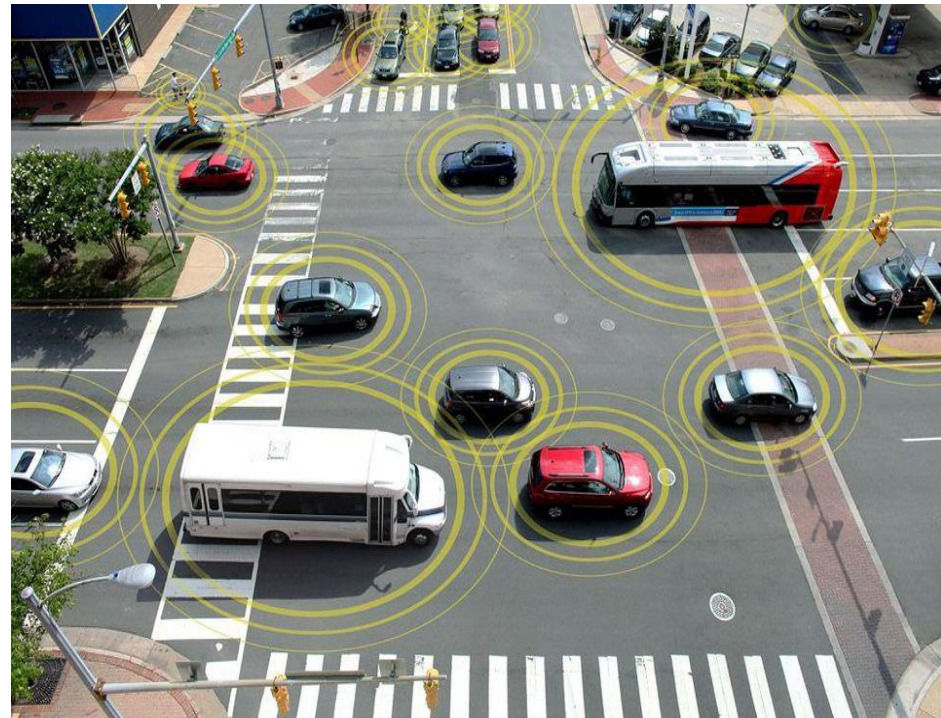
CPYR box Architecture



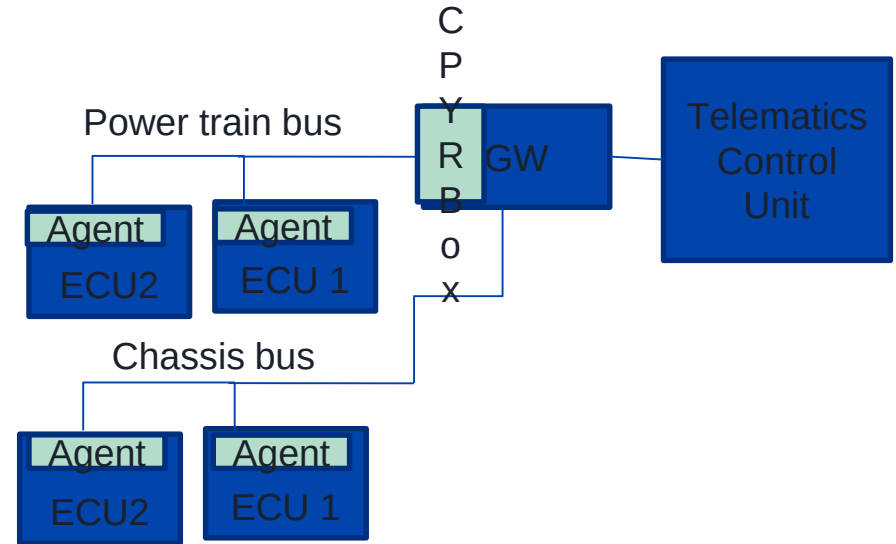
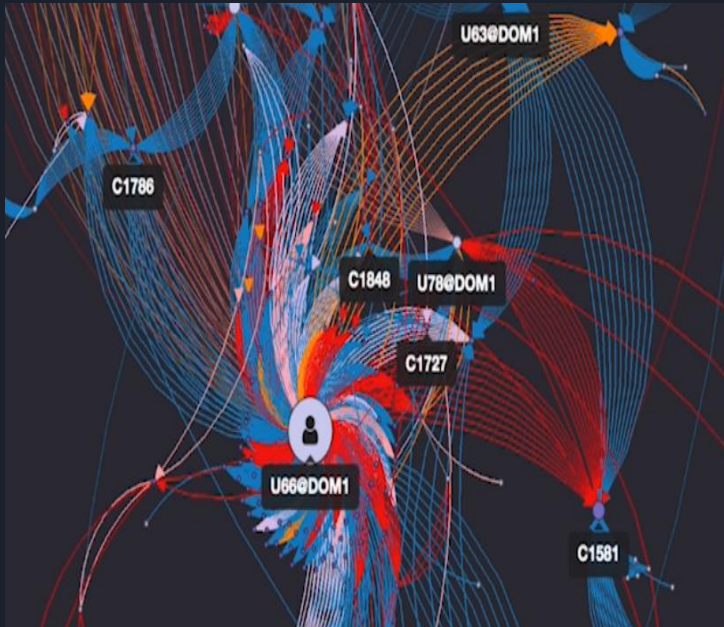
Imaginary Visualization of V2X network



Malicious activity filtration of one vehicle



Imaginary Visualization of activity in one Car

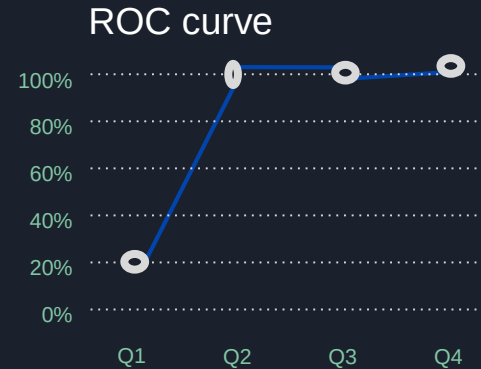


Model evaluation

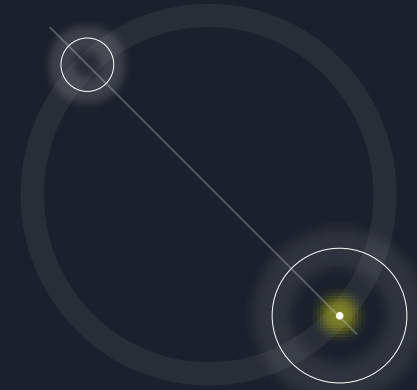
Actual

		Predicted	
		Malicious	benign
Malicious	% TP > 98%	% FN	
benign	% FP	% TN > 98%	

	Sec
Data ingest	
Data Translation	
Feature Engineering	
Model Generation	



- CPYR first Automotive AI Cyber security Product .
- CPYR is using deep learning to learn anomalies and could react in zero time since attack.
- CPYR uses a trusted NVIDIA technology for anomaly detection and compliant Autosar Components.
- CPYR learns during test drives , and requires no additional effort or expenses for deployment.
- Solution could be tailored to work on Gateways or specific safety critical ECUs.



SUMMARY

CPYR your Security watch dog

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